Experiment 0

Familiarization with Synopsys VCS Tool

AIM:

To familiarize with the Synopsys VCS (Verilog Compiler and Simulator) tool flow used for *functional verification of digital designs*. This includes understanding how to:

1. Compile RTL and testbench source files,
2. Simulate the design under test (DUT),
3. Generate and analyze simulation results.

SOFTWARE/TOOLS USED:

Synopsys VCS

THEORY:

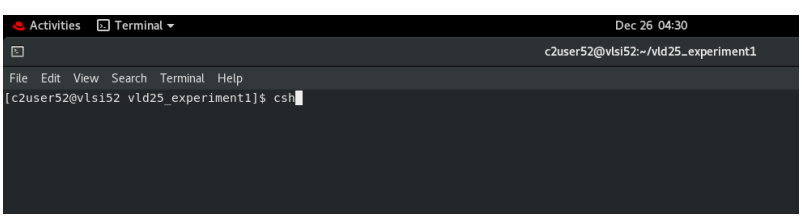
Functional verification is the process of validating that a digital design operates according to its specification before fabrication. This is achieved by simulating the design using appropriate testbenches and analyzing the resulting outputs to detect functional errors at the RTL stage.

Synopsys **VCS (Verilog Compiler and Simulator)** is a high-performance simulation tool used for functional verification of Verilog and SystemVerilog designs. It enables analysis, compilation, and simulation of design descriptions on a single platform and provides facilities for waveform viewing, source-level debugging, and simulation result analysis.

VCS supports all levels of design abstraction and is optimized for **behavioral and register-transfer-level (RTL)** verification. It offers high simulation speed and large design capacity, making it suitable for verifying complex ASIC and SoC designs. Compared to FPGA-oriented tools such as Xilinx Vivado, which are primarily intended for synthesis and implementation, VCS is specifically designed for **industrial-scale functional verification**. Its verification-centric architecture and scalability make it a preferred choice in professional ASIC verification flows.

SIMULATION STEPS USING SYNOPSYS VCS:

Step 1. Launch C Shell
The Synopsys tools in the lab are configured to run under the C shell (csh). The C shell is invoked before starting the verification flow.

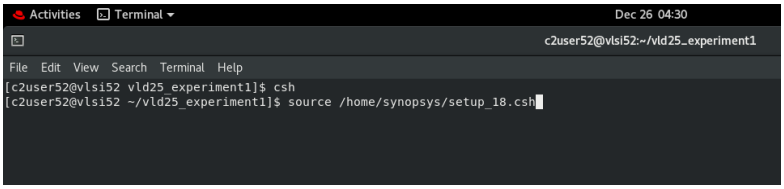


Step 2. Set Up Synopsys Environment
The Synopsys environment is initialized by sourcing the provided setup script.
This step:

- Sets the VCS installation path (VCS_HOME)

- Updates the system PATH
- Configures licensing (LM_LICENSE_FILE)
- Sets compiler and platform dependencies

After this step, the system is ready to invoke VCS commands.

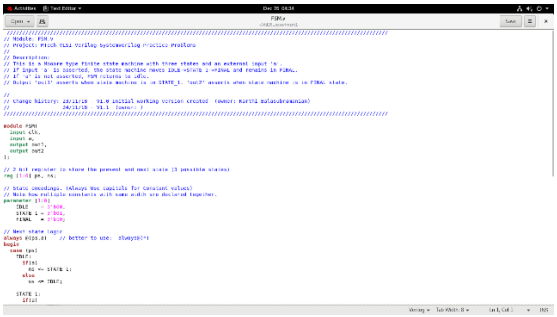


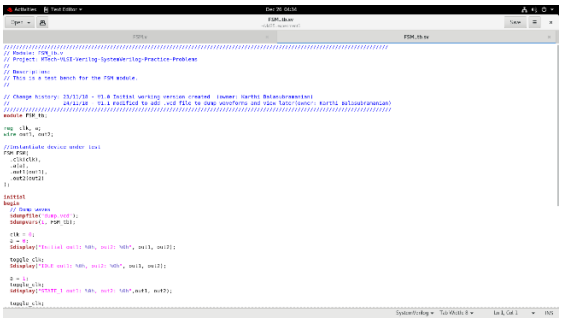
Step 3. Preparation of Source Files

These files are prepared:

- RTL design files (Verilog/SystemVerilog)
- Testbench files

These files together describe the **Design Under Test (DUT)** and its verification environment.





Step 4. Compilation of Simulation Executable

VCS compiles the design and testbench into a native simulation executable.

General syntax:
vcs source_files [source_or_object_files] [options]
Eg: vcs design.v testbench.sv -sverilog

During compilation:

- Source files are parsed and elaborated
- Native object code is generated
- An executable file (simv by default) is created

This compiled-executable approach provides **higher simulation speed and lower memory usage** than interpretive simulators.

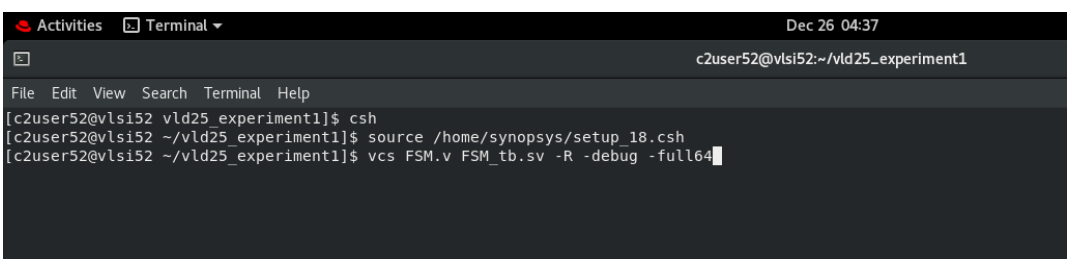
Step 5. Simulation Execution
The compiled executable is run to simulate the design.
General syntax:
./simv runtime_options
Eg: ./simv -l sim.log

Alternatively, compilation and simulation can be combined:
vcs design.v testbench.sv -sverilog -R

Step 6. Enabling Debug and 64-bit Simulation
For effective debugging and support of larger designs, the simulation is typically run with debug and 64-bit options enabled.
Eg: vcs design.v testbench.sv -sverilog -R -debug -full64

- -debug enables source-level debugging and waveform visibility
- -full64 enables 64-bit compilation for improved performance and capacity

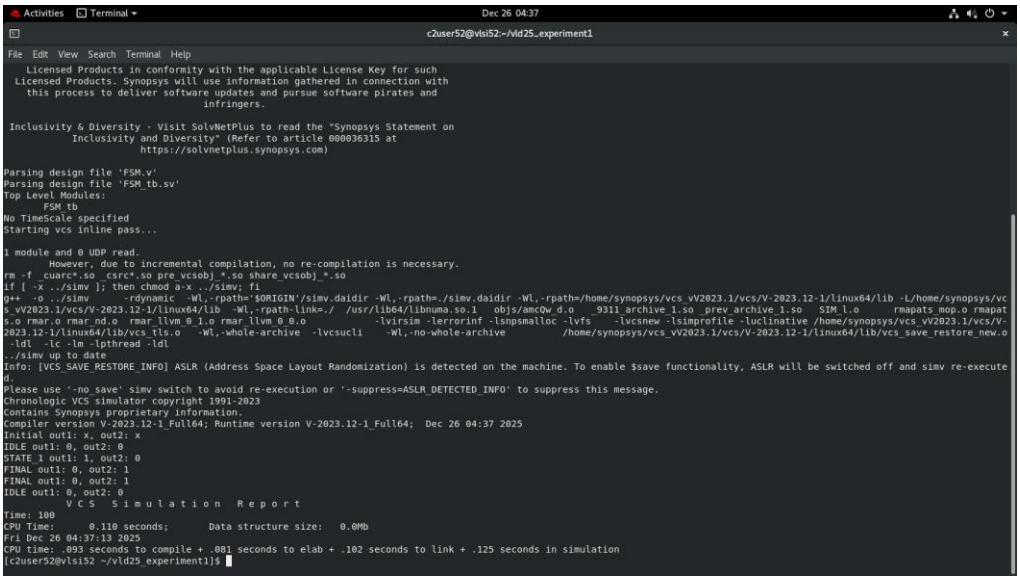
This allows better observation and analysis of simulation behavior.



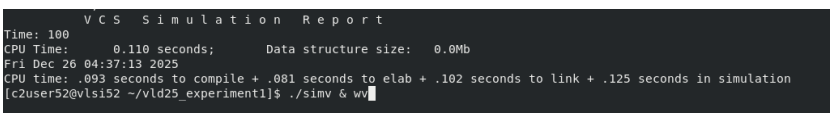
Step 7. Observation of Simulation Results
During simulation execution, the testbench applies stimulus to the Design Under Test (DUT). The simulator produces:

- Console outputs
- Log files
- Debug and waveform information (if enabled)

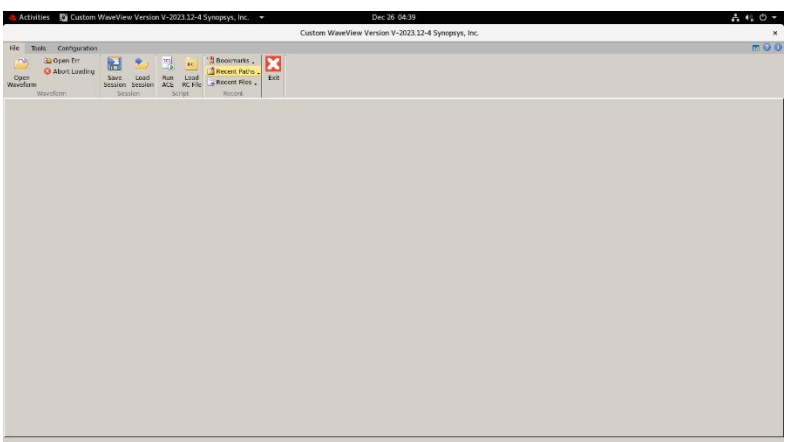
These outputs are examined to verify whether the design behaves as expected.



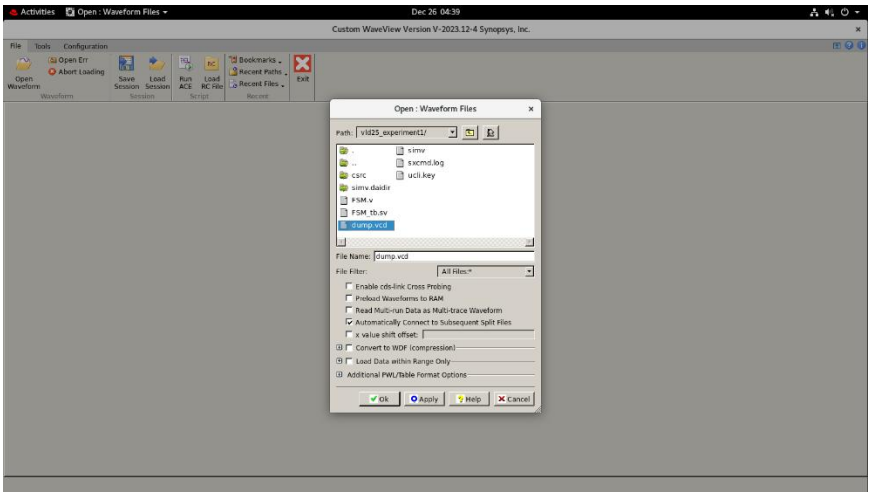
After starting the simulation using the command: ./simv &
the waveform viewer is launched using: wv
This opens the waveform viewing environment to analyze simulation results.



Step 8. Opening the Waveform Window
The waveform window is opened by selecting the **"Open Waveform"** option in the waveform viewer.



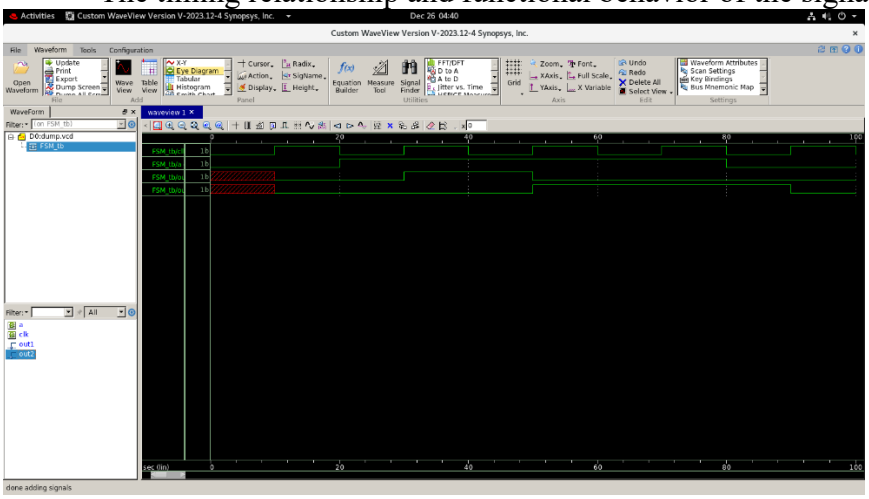
Step 9. Selecting the Simulation Output File (.vcd)
During simulation, a **Value Change Dump (.vcd)** file is generated, which contains signal transition information.
In the waveform viewer, the generated .vcd file is selected and loaded for analysis.



Step 10. Viewing Signals in the Waveform

Once the .vcd file is loaded:

- Input, output, and internal signals are selected
- Signal waveforms are displayed with respect to simulation time
- The timing relationship and functional behavior of the signals are analyzed.



This step helps verify whether the Design Under Test (DUT) operates according to the expected functionality

INFERENCE:

The Synopsys VCS simulation workflow was successfully performed, including environment setup, compilation and execution of SystemVerilog design and testbench files, waveform generation, and waveform analysis. This experiment provides hands-on understanding of RTL functional verification and the use of VCS for analyzing digital design behavior through simulation and waveforms.